

BHAVISH

Full-Stack AI Engineer | LLM Agents & Backend Systems
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SUMMARY

Full-stack AI engineer with 2+ years of experience building autonomous vision-agentic AI pipelines, LLM and RAG systems, real-time streaming architectures (WebSockets, gRPC, SSE), and production desktop/web SaaS across AWS and GCP. Core contributor to two Product Hunt launches: AutoFlow (#1 Product of the Day) and NativeBridge (#3 Product of the Day). Experienced with Git-based version control and CI/CD pipelines (GitHub Actions) for shipping production releases. Known for fast, independent execution: shipped a production VS Code extension, a real-time mobile device streaming farm competing with BrowserStack, and a serverless AI voice SaaS on Modal GPUs.

EXPERIENCE

Full-Stack AI Developer | NativeBridge, Remote

Apr 2025 – Present

Product Hunt #3 Product of the Day • producthunt.com/products/nativebridge-2

- Architected an autonomous vision-agentic UI testing pipeline combining multimodal LLMs with visual grounding for Android devices, translating natural-language prompts (e.g., "complete signup flow") into multi-step mobile test scripts.
- Built the company's official VS Code extension for mobile test automation with fast, independent turnaround, engineering custom token storage and background auto-refresh to work around browser cookie limits; publicly praised by the CEO for delivery speed.
- Designed a low-latency WebSocket + ADB touch-gesture relay for remote Android/iOS physical devices, powering an interactive real-time touch-canvas streaming UI as part of a BrowserStack-style device farm.
- Built an end-to-end telemetry pipeline (on-device gRPC server → backend provider → main backend → SSE stream) delivering live device XML and screenshots for real-time Maestro code generation.
- Integrated an Appium Inspector proxy and architected a Self-Healing Test System (Python/FastAPI) that intercepts failed element lookups and auto-recovers broken locators via DOM-tree similarity search.
- Managed version control with Git/GitHub and used GitHub Actions to automate build, test, and deployment (CI/CD) for production releases.

Full-Stack AI Developer | AutoFlow, Remote

Sep 2024 – Apr 2025

Product Hunt #1 Product of the Day • producthunt.com/products/autoflow-studio

- Architected an autonomous vision-agentic UI testing pipeline for browser automation, combining multimodal LLMs with visual grounding to generate multi-step Playwright scripts from natural-language prompts (the browser-focused counterpart to the mobile version later built at NativeBridge).
- Designed Shadow DOM locator-extraction algorithms and an interactive multi-locator editing UI within the Electron desktop app.
- Delivered onboarding flows and a scalable component library on a tight pre-launch timeline, helping scale the platform for its Product Hunt #1 debut.
- Led frontend desktop architecture using React, MobX, and Electron.js IPC, with Git for collaborative version control across the team.

Frontend Developer Intern | AutoFlow, Remote

Dec 2023 – Sep 2024

- Built a high-precision element-inspection tooltip for live browser views, shared directly by the CEO and praised by clients.
- Developed hands-on expertise in Electron.js IPC, React component architecture, and MobX state management across complex web and desktop interfaces.

PROJECTS

JobPilot | AI Job-Search & Application-Prep Platform

github.com/Bhavish77/jobpilot

Tech: Python, FastAPI, Celery, LangGraph, Node.js, Next.js 15, PostgreSQL, Redis, RabbitMQ, Google Gemini API

- Designed a distributed, atomic rate-limiting system in Redis (a lazily refilled Lua-script token bucket plus a separate daily-quota gate) protecting a shared LLM budget across a Celery worker pool, with each check-and-consume step atomic under concurrency.
- Implemented Postgres-checkpointed pipeline state with human-in-the-loop approval and automatic recovery from transient LLM provider failures; a retry resumes at the exact step that failed, never from scratch.
- Architected a polyglot multi-service system (FastAPI, Celery, Node.js, Next.js) built around a resumable LangGraph multi-agent pipeline that drafts tailored resumes, cover letters, and interview prep grounded in the user's real background; deliberately prepare-only, it never auto-submits on the user's behalf.

- Built an automated ATS-verification loop that compiles generated resumes to real PDFs, extracts the text layer the way an ATS parser would to verify formatting survives, and rewrites content that overflows a page limit by relevance.
- Shipped the full Next.js 15 frontend with a custom accessible component library and real-time status updates pushed over a JWT-authenticated WebSocket service.

AskDocs | RAG Document Q&A Platform

github.com/Bhavish77/chatpdf · askdocs-e8m6.onrender.com

Tech: Python 3.12, FastAPI, LangGraph, Google Gemini API (generation + embeddings), PostgreSQL 16, pgvector (vector search), psycpg3, SSE, Docker, GitHub Actions, Render, Neon

- Built a self-correcting RAG chat pipeline with LangGraph over a pgvector vector database with Gemini embeddings (retrieve → grade → conditional rewrite → generate → verify) that grades its own retrieval quality and checks final answers for groundedness before returning them.
- Designed a crash-safe async document ingestion system using a hand-written PostgreSQL job queue (FOR UPDATE SKIP LOCKED) with exponential backoff, dead-lettering, and idempotent upserts; verified live that killing the worker mid-job and restarting produces zero duplicate data.
- Implemented authentication and per-user data isolation from scratch (argon2id hashing, server-side sessions, CSRF protection, login throttling, rate limits), with an automated test suite proving one user cannot access another's documents, chats, or files.
- Engineered per-document-balanced vector search (PostgreSQL window functions over pgvector) ensuring fair representation across multiple documents tagged in a single query.
- Built a token-by-token streaming chat UI (Server-Sent Events) with live citation rendering and a step-by-step execution trace, in vanilla JavaScript with no frontend framework.
- Wrote 64 automated tests covering auth, data isolation, async job mechanics, and the LLM pipeline (using a fake LLM client), running in GitHub Actions CI against a real Postgres instance on every push.
- Diagnosed and fixed subtle bugs found only through live testing against the real Gemini API (an embedding-batching mismatch, a pgvector type-casting issue, and a CSP policy gap).

Resonance | Full-Stack AI Voice-Generation SaaS

resonance-kappa-seven.vercel.app

- Built a multi-tenant AI voice-generation SaaS hosting the open-source Chatterbox TTS model on Modal serverless GPUs.
- Shipped a FastAPI backend and React frontend with Paddle subscription billing, multi-tenant workspace management, AWS S3 audio storage, and Google OAuth.

TECHNICAL SKILLS

AI & Multimodal Agents: LLM Agents, Multi-Agent Systems, RAG (Retrieval-Augmented Generation), Vision LLMs, LangChain, LangGraph, Google Gemini API, Prompt Engineering, Claude Computer Use API, Chatterbox TTS, Modal Serverless GPUs, Vercel AI SDK

Full-Stack: TypeScript, JavaScript, Python, C++, React.js, Next.js 15, Node.js, FastAPI, Electron.js, MobX, Tailwind CSS, Radix UI

Backend Infrastructure & Messaging: Celery, RabbitMQ, Redis, SQLAlchemy (async), Alembic, Distributed Rate Limiting, Pub/Sub Architecture, Lua Scripting

Real-Time & Streaming: WebSockets, gRPC, Server-Sent Events (SSE), RESTful APIs, Nginx

Databases: PostgreSQL, SQL, MongoDB, NoSQL, Vector Databases (pgvector, Pinecone, Qdrant, Chroma), Vector Search, Embeddings

Cloud, DevOps & Version Control: AWS (S3), GCP, Docker, Docker Compose, Git, GitHub, GitHub Actions, CI/CD, Render, Vercel, Google OAuth 2.0, JWT, Paddle Billing

Testing & Automation: pytest, pytest-asyncio, Appium, Playwright, Maestro, ADB, DOM Similarity Search, Self-Healing Proxies

EDUCATION

M.Sc. Computer Science

Graduated 2024